BECE407P ASIC DESIGN

Module - 6: Physical Design

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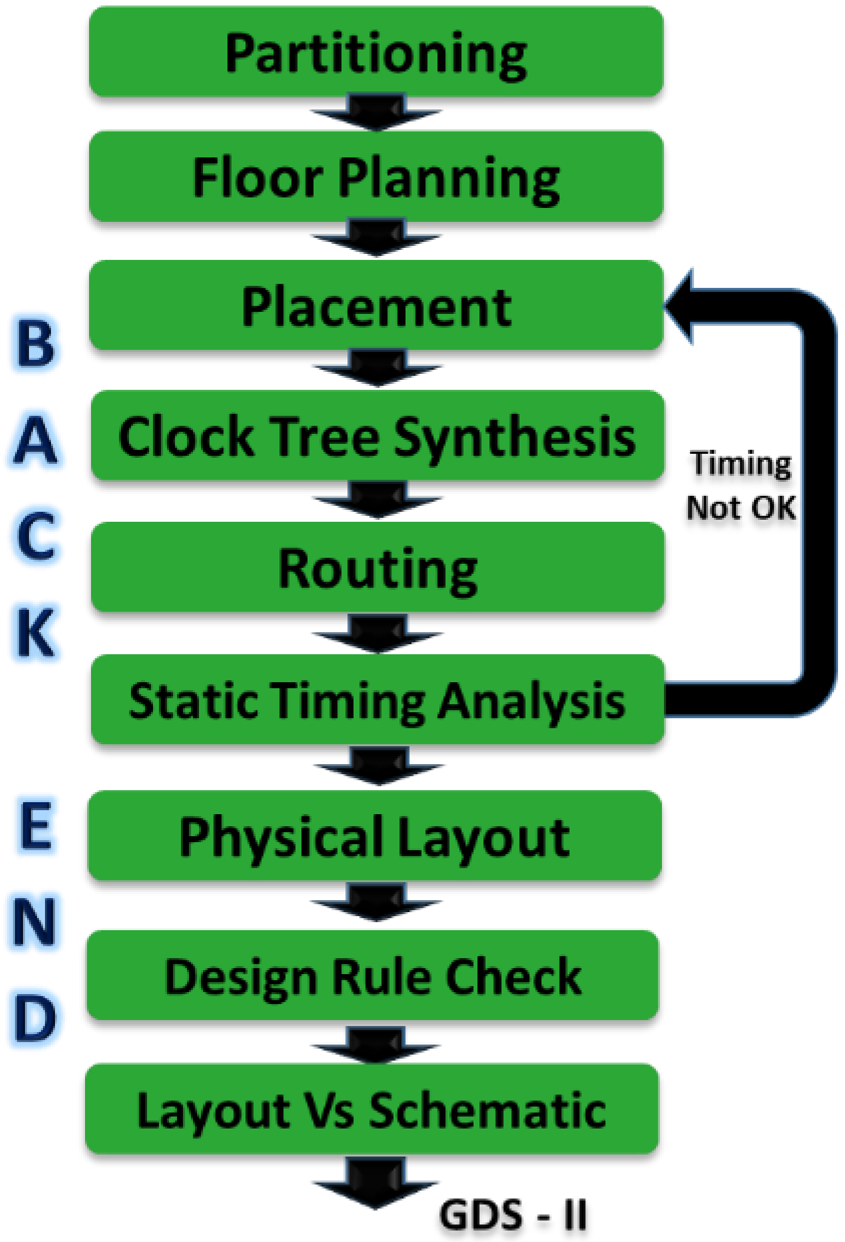
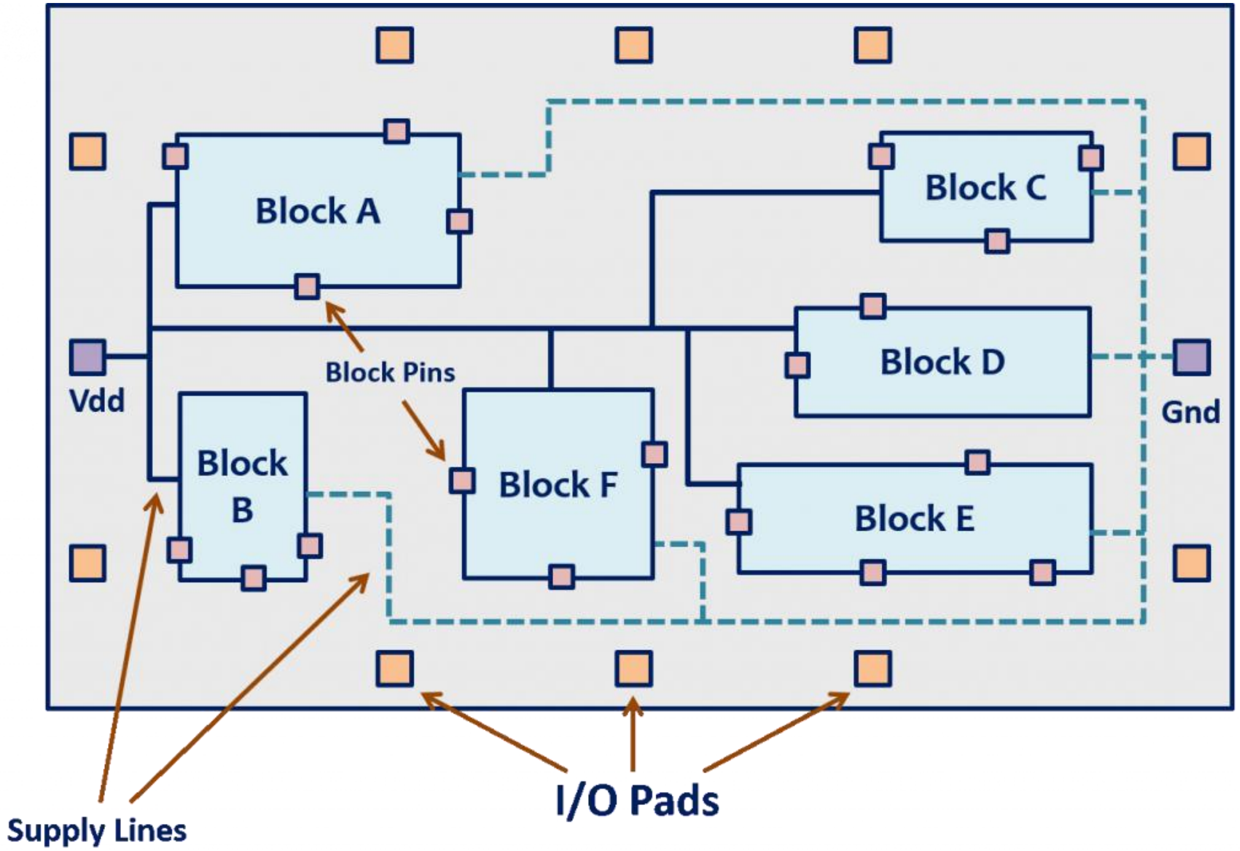
  - ❑ Clock Tree Synthesis (CTS)
  - ❑ Routing
  - ❑ Parasitic Extraction
  - ❑ Physical Verification (DRC/LVS)
  - ❑ ECO/Tapeout



Partitioning & Floorplanning

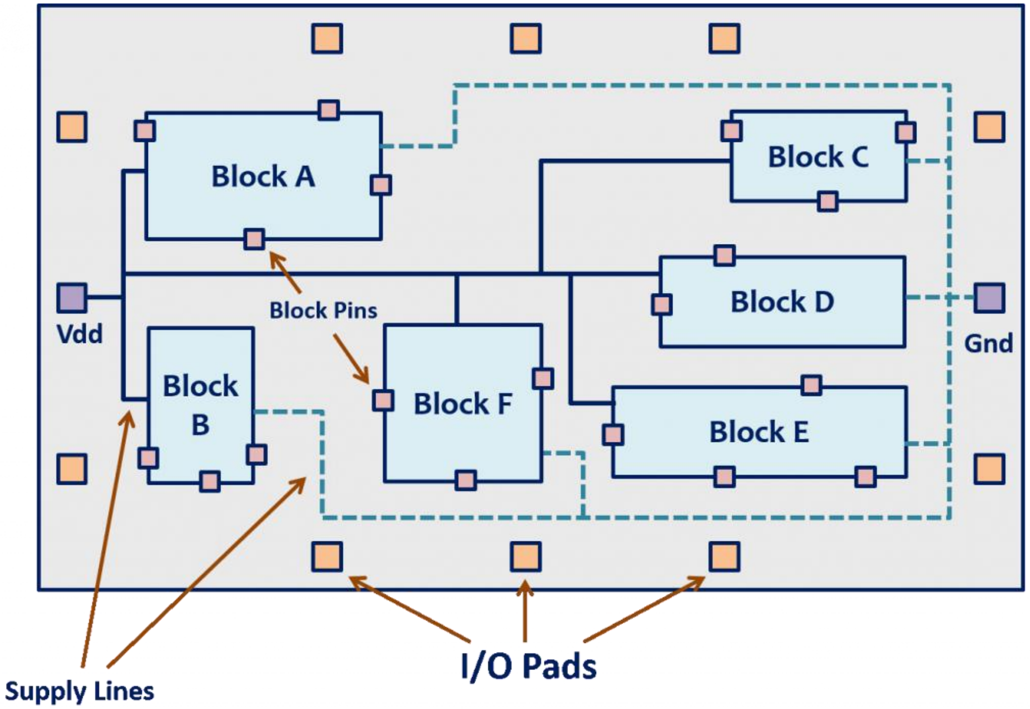
Partitioning: It involves the decomposition of a complex circuit into smaller subsystems (sub-blocks) so that each subsystem will be of a manageable size and the no. of interconnections between these sub-blocks will be minimum.

Floor planning: In this step, the rough position of each block on the silicon chip and the shape of each block will be decided. And also, the pin locations of each block will be decided so that the interconnections between these blocks can be routed easily in the future as shown in the below picture.



Partitioning & Floorplanning

- Purpose: Define chip architecture and spatial layout for macros and standard cells.
- Inputs: Netlist, *.lef* files, utilization target, aspect ratio.



Key Activities:

- Define die/core boundary and aspect ratio.
- Place I/O pads, macros, hard IPs.
- Define keep-out zones, placement blockages.
- Create power rings and block-level channels.
- Evaluate congestion map and area utilization.

Output: Floorplan DEF file with macro locations, IO pin placements, and power ring locations.

Power Planning & Placement

Power planning sets up robust power distribution to all logic cells.

Types:

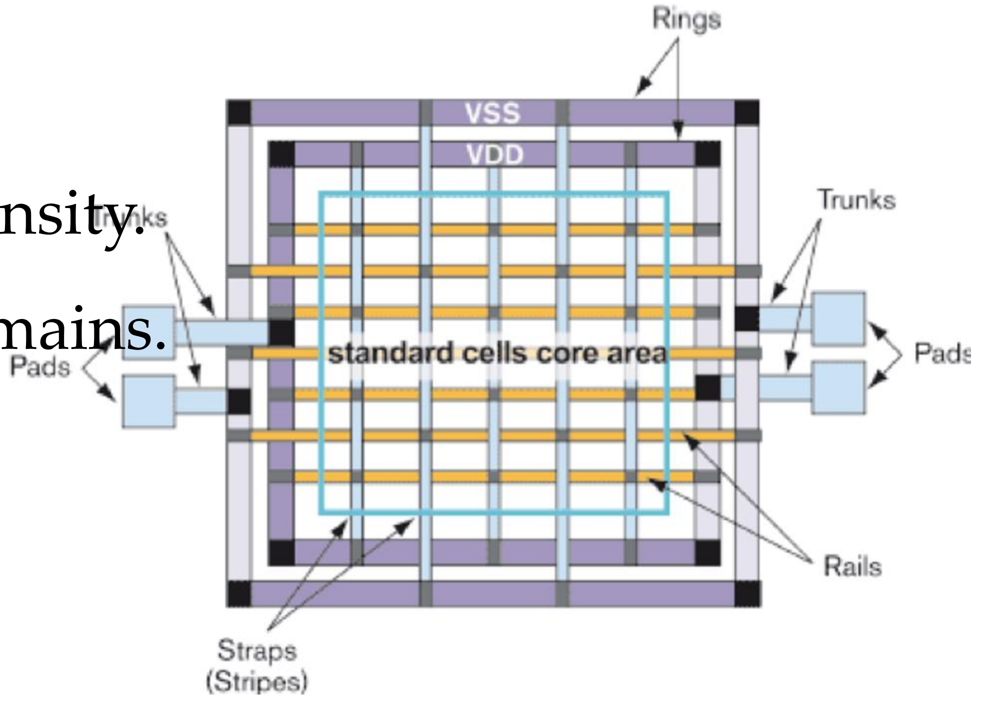
Core cell power management: Core ring (VDD/VSS) and vertical/horizontal straps.

IO cell power management: Power path from IO pads to core rings.

Considerations:

- IR drop and electromigration control.
- Width/spacing of power strips as per current density.
- Separate supplies for analog, digital, and IO domains.

Tools: Innovus, ICC2, or Voltus PowerGrid Analysis.



Placement

Objective: Position standard cells optimally for timing and routability.

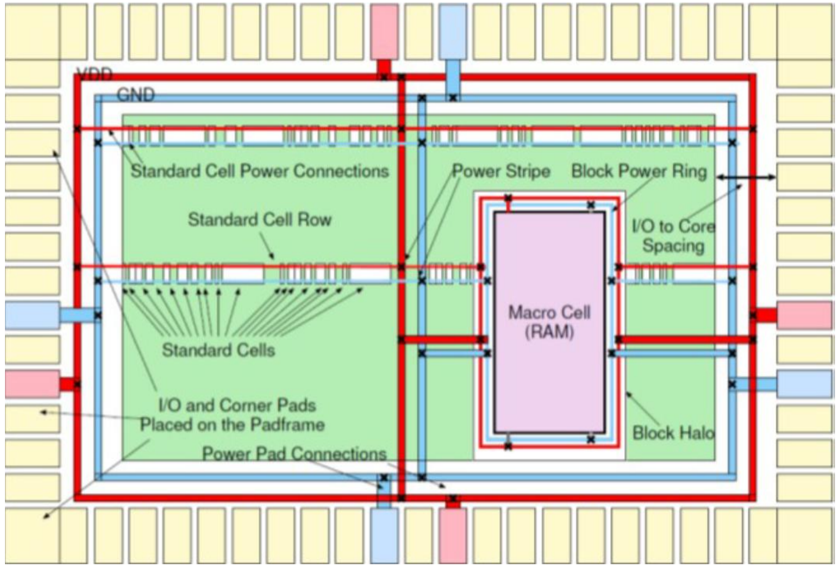
Steps:

1. Pre-placement setup (define regions, cell density targets).
2. Initial placement (timing-driven or congestion-driven).
3. Legalization (remove overlaps, satisfy cell site alignment).
4. Optimization (insert buffers, resize gates for timing).

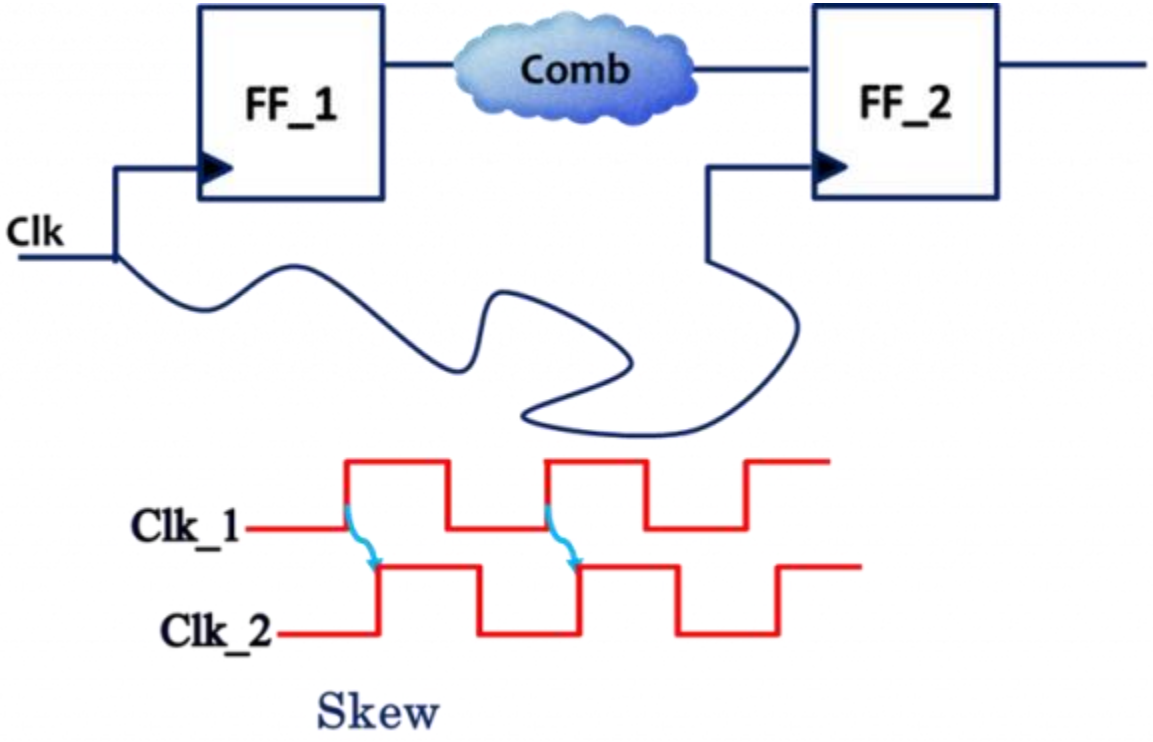
Outputs: Placed DEF file, timing/congestion reports.

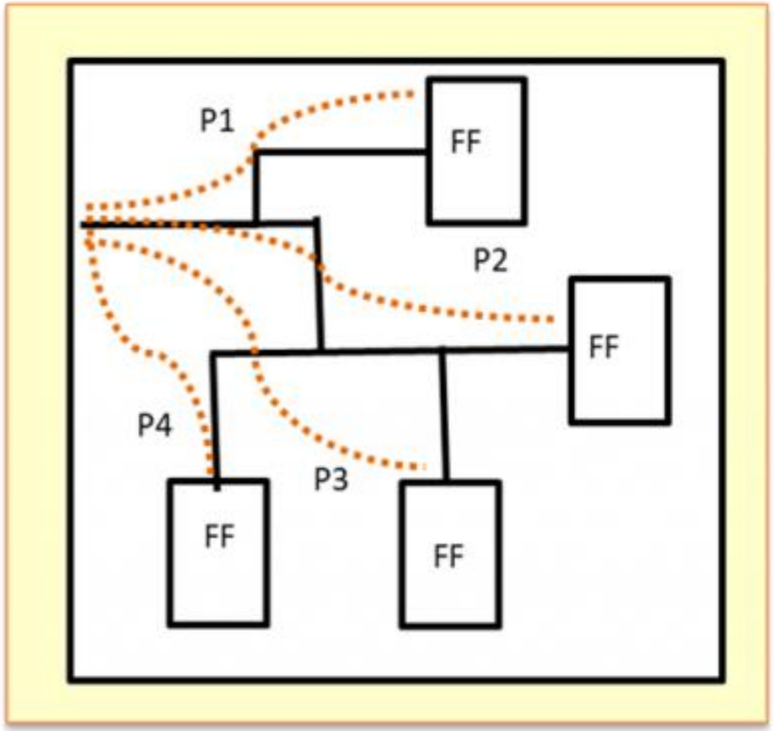
Goals:

- Minimize total wirelength.
- Balance path delays.
- Reduce congestion and DRCs.



Clock Tree Synthesis (CTS)

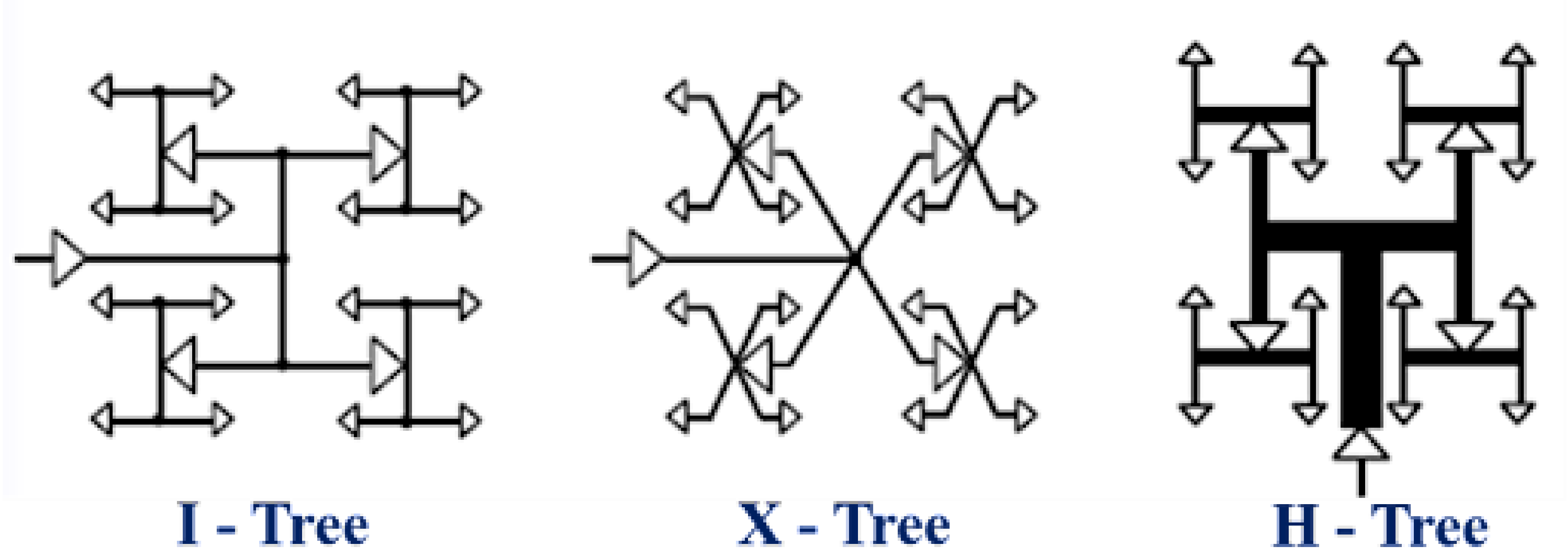




After the completion of floor planning and placement, millions of flip-flops in the complex circuit will be scattered at multiple places on the silicon chip, so that the transmission delays of the clock signal for all the flip-flops will not be the same. The clock will not arrive/reach all the flip-flops at the same time as shown in the figure below. The difference in the arrival times of the clock signal to these flip-flops is referred to as *clock skew*.

Clock Tree Synthesis (CTS)

It is mandatory to make sure that the clock signal reaches multiple flip-flops with no or minimum skew difference. CTS will ensure that the clock reaches various flip-flops almost at the same time. In the clock tree synthesis process, any one of the standard Tree topology structures (shown in the below picture) will be implemented in the design with an intention of making the length of all the wires in the clock distribution network equal.



Clock Tree Synthesis (CTS)

Purpose: Distribute clock with controlled skew and latency.

Inputs: Post-placement netlist and parasitic.

CTS Stages:

1. Clock Net Identification
2. Clock Buffer Insertion
3. Clock Balancing and Optimization
4. Post-CTS Hold Fixing

Common Topologies: H-Tree, X-Tree, Fishbone (I-Tree).

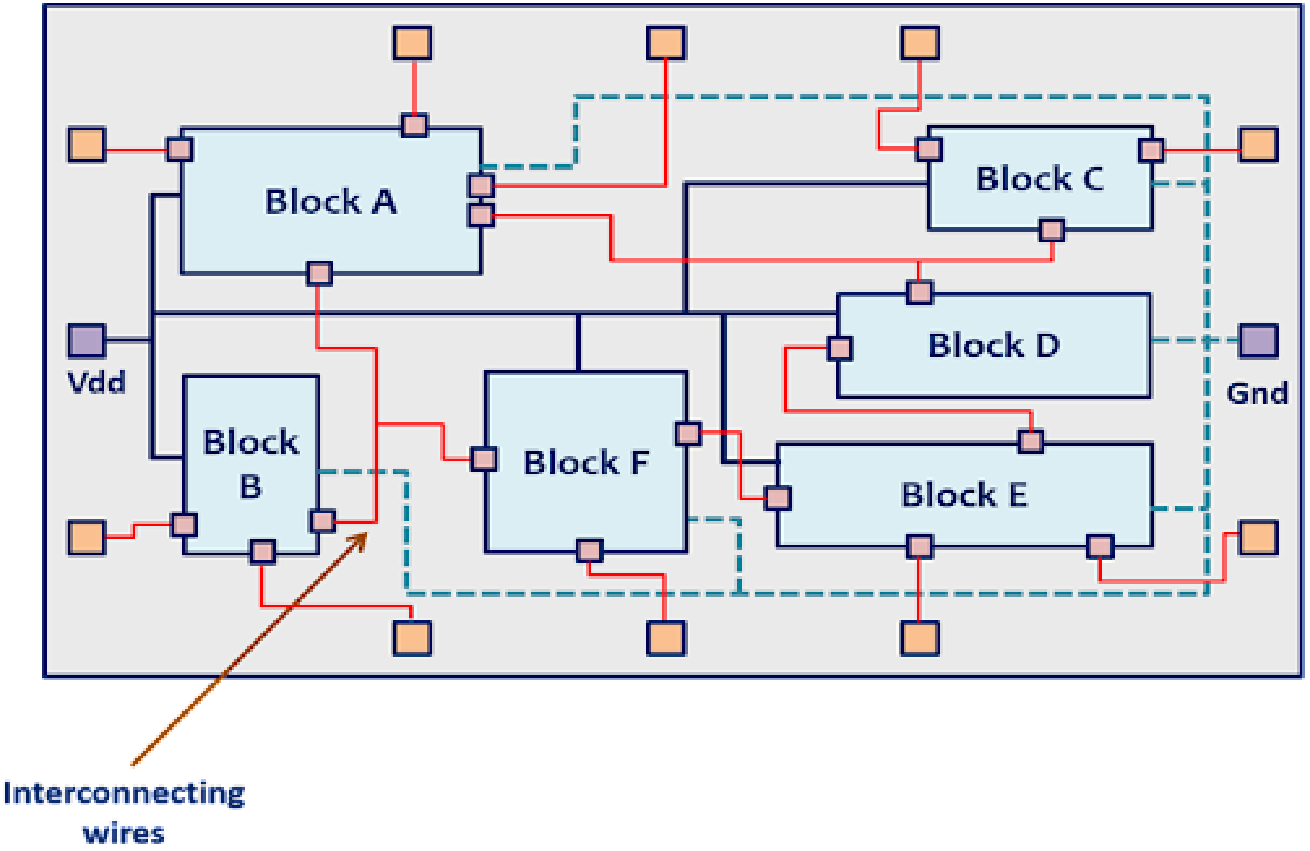
Outputs:

- Balanced clock network.
- Skew, latency, insertion delay reports.

CTS Checks: Ensure no hold/setup violations, no unnecessary transition degradation.

Routing

In this step, interconnecting wires between various blocks will be routed in such a way that, the length of the interconnecting wires will be minimum for meeting the timing requirements and for ensuring that, the chip area will not be increased as shown in the below picture. Proper routing will also ensure that there are no congestion hotspots in the chip so that, the probability of having faults during the fabrication will be less.



Routing

Objective: Establish physical connections between pins and cells following metal rules.

Stages:

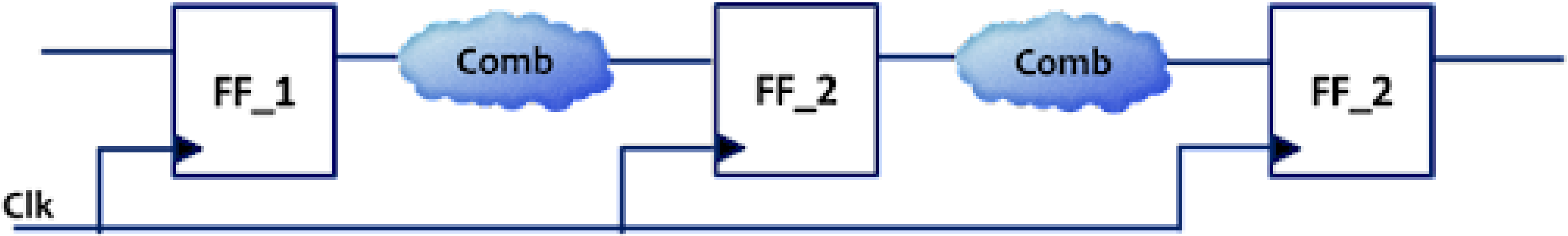
1. *Global Routing*: Estimate routes at a coarse level; guides detailed router.
2. *Track Assignment*: Assign routing tracks and layers.
3. *Detailed Routing*: Generates actual wire paths with DRC compliance.
4. *Search & Repair*: Fix congestion, shorts, opens, antenna violations.

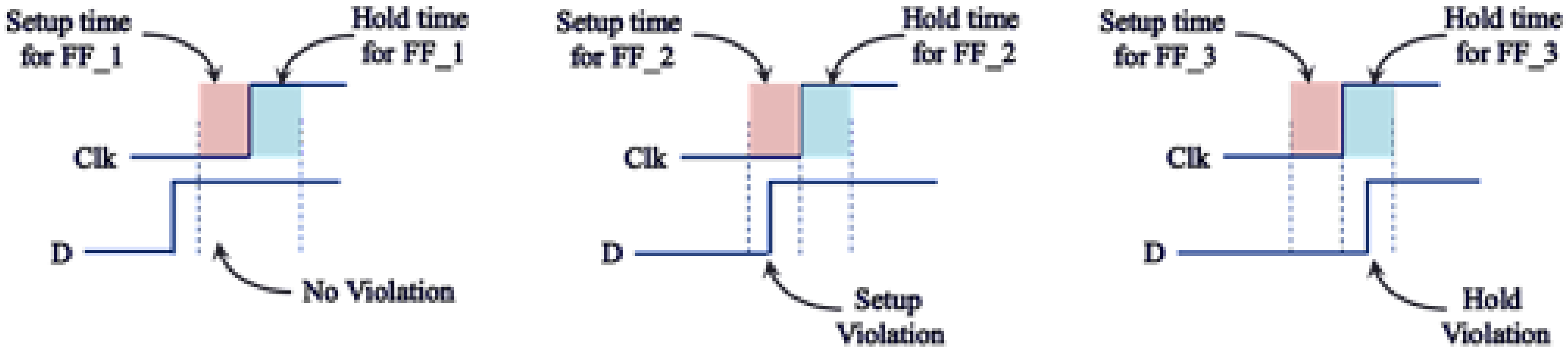
Post-Routing Checks: Verify DRC, IR drop, timing, and electromigration.

Output: Routed DEF, *.spef* files for extraction.

Static Timing Analysis

Static Timing Analysis will be done on the final routed netlist which will represent/mimic the final layout of the chip that is going to be fabricated for validating the timing performance of the chip and to ensure that, the chip will meet the timing requirements such as maximum frequency of operation and the design works fine at the desired frequency without any setup and hold time violations.

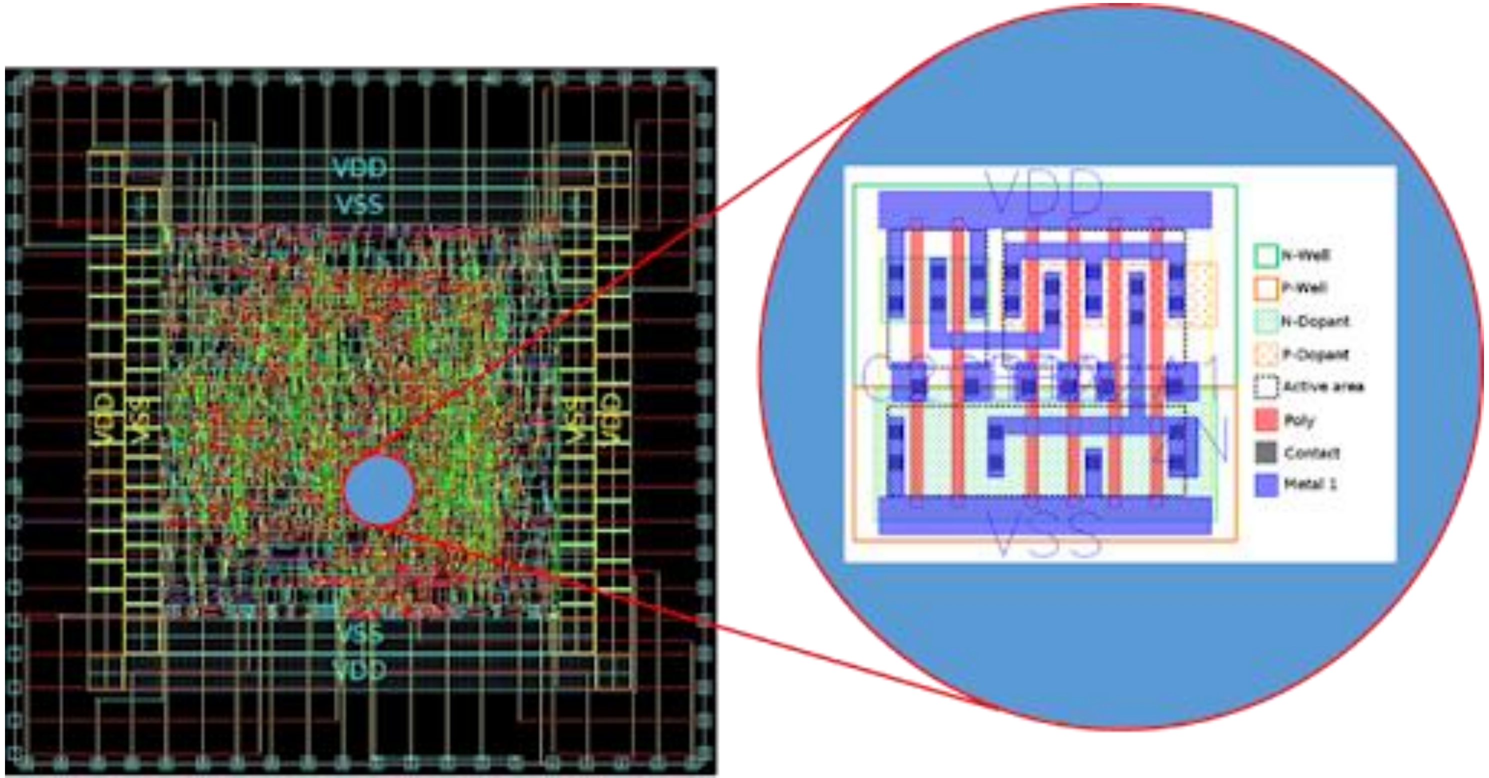




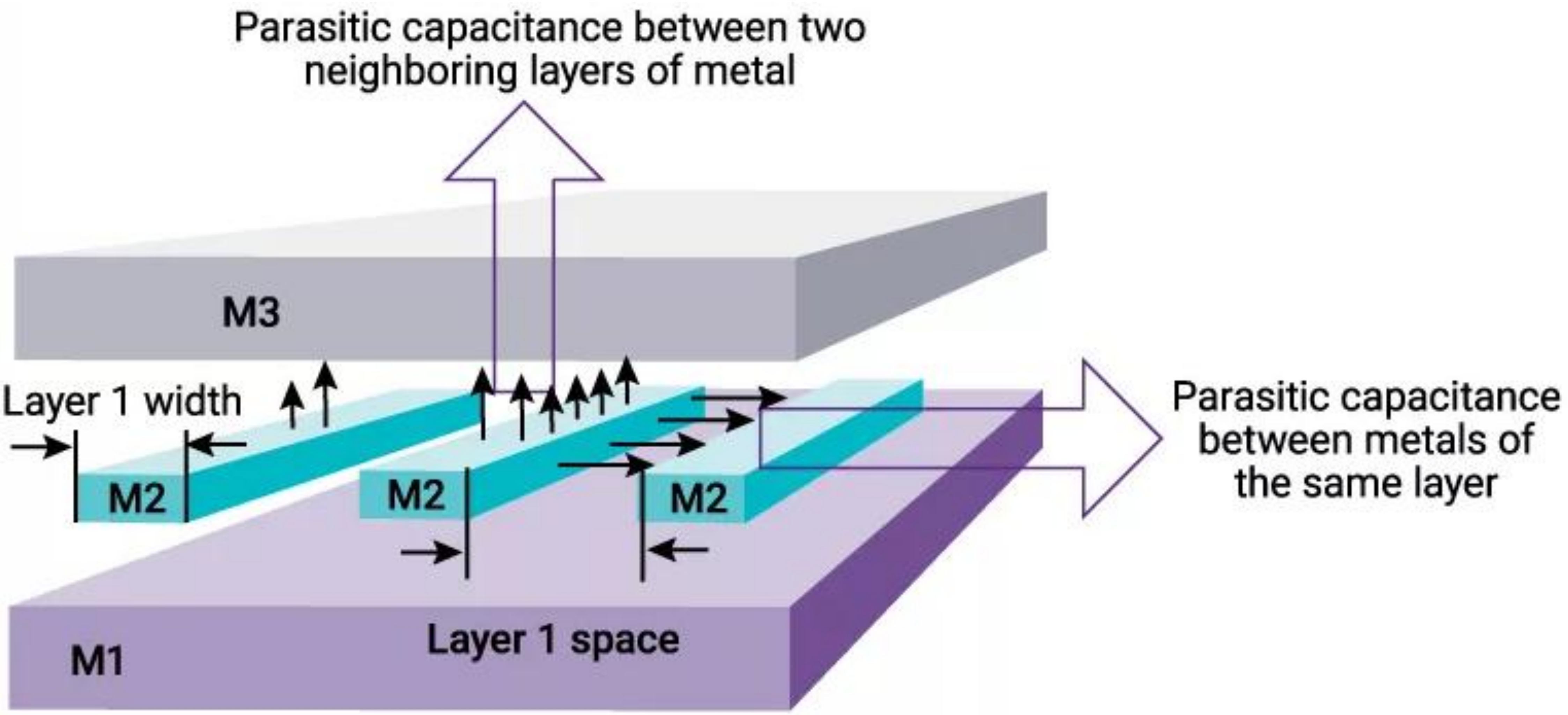
Physical Layout

If the timing requirements are not met or if there are any setup and hold time violations identified, then, optimizing the placement and routing will be done, so that, timing may be improved, and violations will be eliminated.

Post the timing requirements are satisfied, **the physical layout as shown below will be extracted**, and these layouts will be used during the fabrication to prepare the required mask layers which will be used in the photolithography processes.



Parasitic Extraction (PEX)



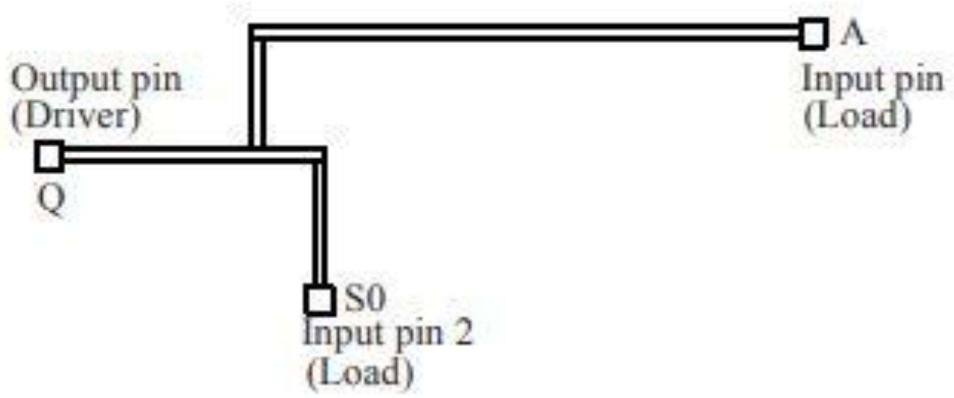
Parasitic Extraction (PEX)

Standard Parasitic Extraction Format (SPEF) allows the representation of parasitic information of a design (R, L, and C) in an ASCII (American Standard Code for Information Interchange exchange format). A user can read and check the values in a SPEF file. **Users would never create this file manually it is automatically generated by the tool**. **It is mainly used to pass parasitic information from one tool to another**. Interconnect parasitic depends on the process. SPEF supports the specification of all the cases like best-case, typical, and worst-case values. These triplets (best, typical, and worst) are allowed for R, L, and C values, ports slows, and loads. The units of the parasitic R, C, and inductance L are specified at the beginning of the SPEF file.

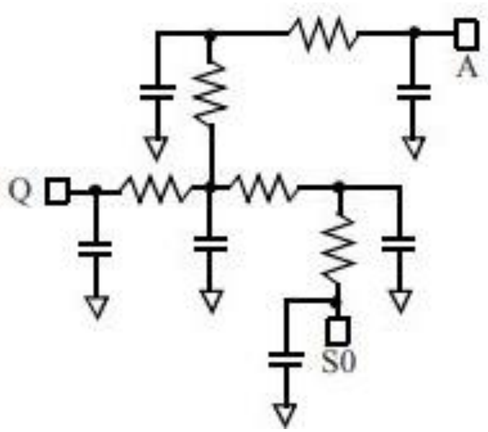
Parasitic Extraction (PEX)

SPEF can be generated by PnR tool or a parasitic extraction tool, and then this SPEF is used by timing analysis tool for checking the timing, in-circuit simulation or to perform crosstalk analysis.

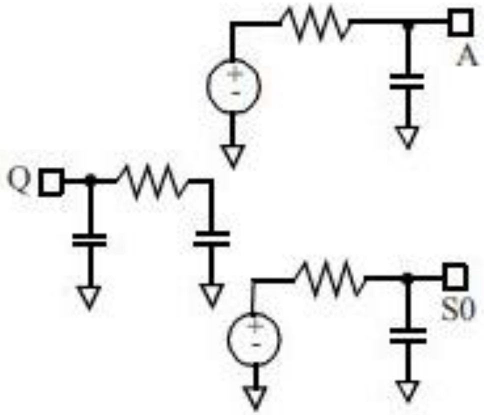
Parasitic can be represented at many different levels. SPEF supports three models.



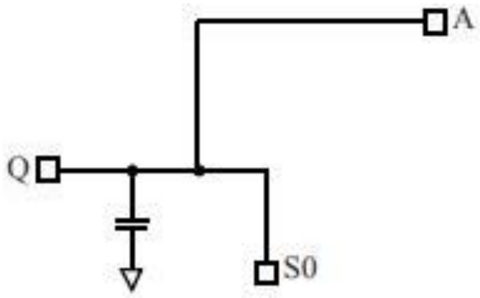
Parasitic Extraction (PEX)



Distributed net model: In this model (D_NET), each segment of a net route has its own R and C values.



Reduced net model: In this model (R_NET), on the load pins of the net have single reduced R and C, and on the driver pin of the net a pie model (C-R-C) is considered.



Lumped capacitance model: In this model, only a single capacitance is specified for the entire net.

Parasitic Extraction (PEX)

Parasitic extracted from a layout can be defined in three formats:

Detailed Standard Parasitic Format (DSPF)

Reduced Standard Parasitic Format (RSPF)

Standard Parasitic Extraction Format (SPEF)

The SPEF is a compact format of the detailed parasitic. A SPEF file for a design can be split across multiple files and it can also be hierarchical. SPEF is the format of choice for representing the parasitic in a design due to its compactness and completeness

Parasitic Extraction (PEX)

An example of a net with two fanouts (*I *8 and *I *10) is given below.

***D_NET** NET_27 0.77181
***CONN**
*I *8: Q O *L 0 *D CELL1
*I *10: I I*L 12.3
***CAP**
1 *9:0 0.00372945
2 *9:1 0.0206066
3 *9:2 0.035503
4 *9:3 0.0186259
5 *9:4 0.0117878
6 *9:5 0.0189788
7 *9:6 0.0194256
8 *9:7 0.0122347
9 *9:8 0.00972101
10 *9:9 0.298681
11 *9:10 0.305738
12 *9:11 0.0167775
***RES**
1 *9:0 *9:1 0.0327394
2 *9:1 *9:2 0.116926
3 *9:2 *9:3 0.119265
4 *9:4 *9:5 0.0122066
5 *9:5 *9:6 0.0122066
6 *9:6 *9:7 0.0122066
7 *9:8 *9:9 0.142205
8 *9:9 *9:10 3.85904
9 *9:10 *9:11 0.142205
10 *9:12 *9:2 1.33151
11 *9:13 *9:6 1.33151
12 *9:1 *9:9 1.33151
13 *9:5 *9:10 1.33151
14 *9:12 *8: Q 0
15 *9:13 *10: I 0
***END**

<u>CONTENTS in SPEF FILE</u>:
header_definition: contains basic information such as the SPEF version number, design name, and units for R, L and C.
[name_map]: specifies the mapping of net names and instance names to indices.
[power_definition]: declares the power nets and ground nets.
[external_definition]: defines the ports of the design.
[define_definition]: identifies instances, whose SPEF is described in additional files
internal_definition: contains the guts of the file, which are the parasitic of the design.

Parasitic Extraction (PEX)

Purpose: Estimate *capacitance* (C), *resistance* (R), and *inductance* (L) for all nets.

Tools: Synopsys StarRC, Cadence QRC, Siemen's MentorGraphics Calibre.

Output Formats: SPEF (*Standard Parasitic Exchange Format*), RSPF.

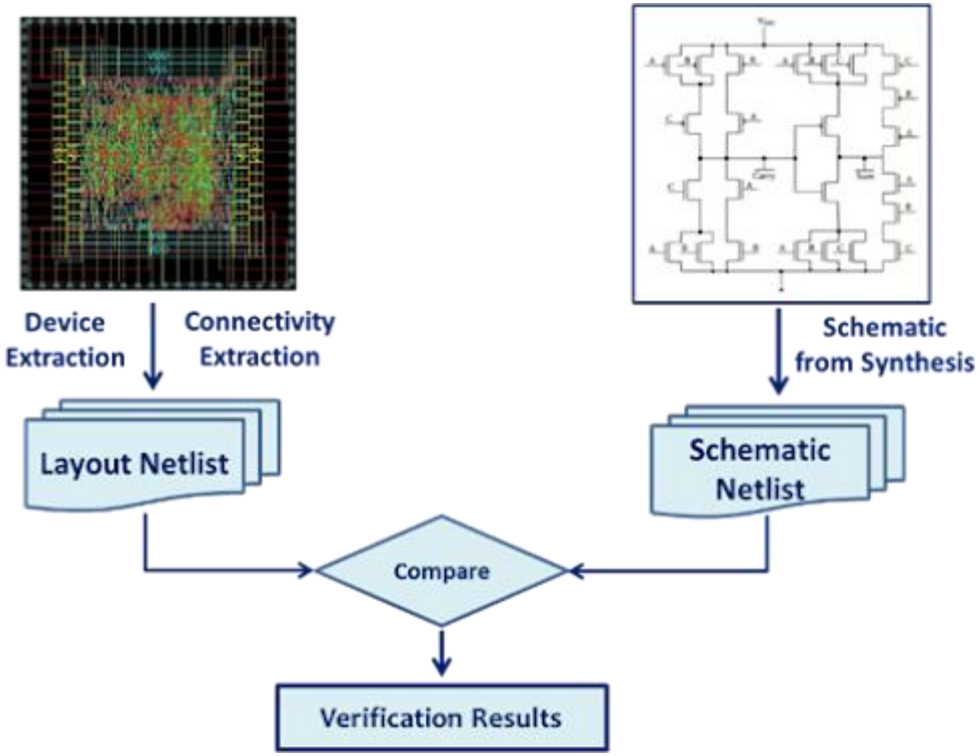
Significance:

- Back-annotate experience into post-layout timing (STA).
- Identify coupling capacitance for signal integrity checks.

Physical Verification (DRC/LVS)

Design Rule Check: The physical layout must obey the design rules imposed by the foundry to ensure that, the layout is manufacturable with a minimum no. of fabrication faults. They act as an interface or communication link between the circuit designer and the process engineer during the manufacturing phase. So, to ensure that the layout obeys all the design rules, a DRC will be done.

Layout Vs Schematic: Finally, to ensure that the design's functionality is not altered during the physical design implementation Layout, the Layout Vs Schematic (LVS) equivalence checking as shown in the below picture will be done. This equivalence as shown below will be able to identify if there are missing device terminals or extra device terminals or unmatched devices and nets in the layout/schematic.



Physical Verification (DRC/LVS)

Design Rule Check (DRC): Verifies geometries comply with foundry rules.

Layout vs Schematics (LVS): Ensures logical connectivity matches schematic/netlist.

Tools: Calibre, Pegasus, ICV.

Outputs: DRC clean, LVS matched reports.

Once this implementation flow which includes floor planning, placement, CTS, Routing, and Signoff flow with DRC and LVS is completed, the final GDS-II file will be sent to the foundry where the actual silicon chips are created through a fabrication process using Silicon wafer.

Guidelines for Floorplan

- Define die/core size and aspect ratio (ideally 1:1 for balanced routing).
- Place and fix macros and I/O pins carefully, avoiding notches or unnecessary gaps.
- Reserve regions for standard cells, power mesh, and blockages.
- Consider hierarchical (block-based) vs. flat floorplans, as hierarchical designs enable team partitioning.
- Add special cells (tap, end cap) to prevent latch-up and edge issues.
- Output of floorplanning includes placed macros, pin assignments, and blocked regions.

Guidelines for Placement

- Position standard cells optimally for timing, congestion, and area utilization.
- Follow stages: pre-placement, initial placement, legalization, buffering.
- Legalization ensures physical design rules and manufacturability are met, avoiding overlaps.
- Insert spare cells for future ECOs and special protection cells (antenna diodes, well taps).
- Key aims: minimize wirelength, congestion hotspots, and improve timing closure.

Guidelines for Clock Tree Synthesis (CTS)

- Balance clock skew and minimize insertion delay using buffers and inverters.
- Common structures: H-tree, Fishbone, X-tree, multi-level.
- CTS includes clustering, balancing, clock routing, and post-conditioning (fixing skew, cleaning up after routing).
- Optimize by buffer sizing, gate sizing, and dedicated high-fanout net synthesis.
- Final checks: ensure clock latency and skew within specifications, optimize power and area.

Guidelines for Routing

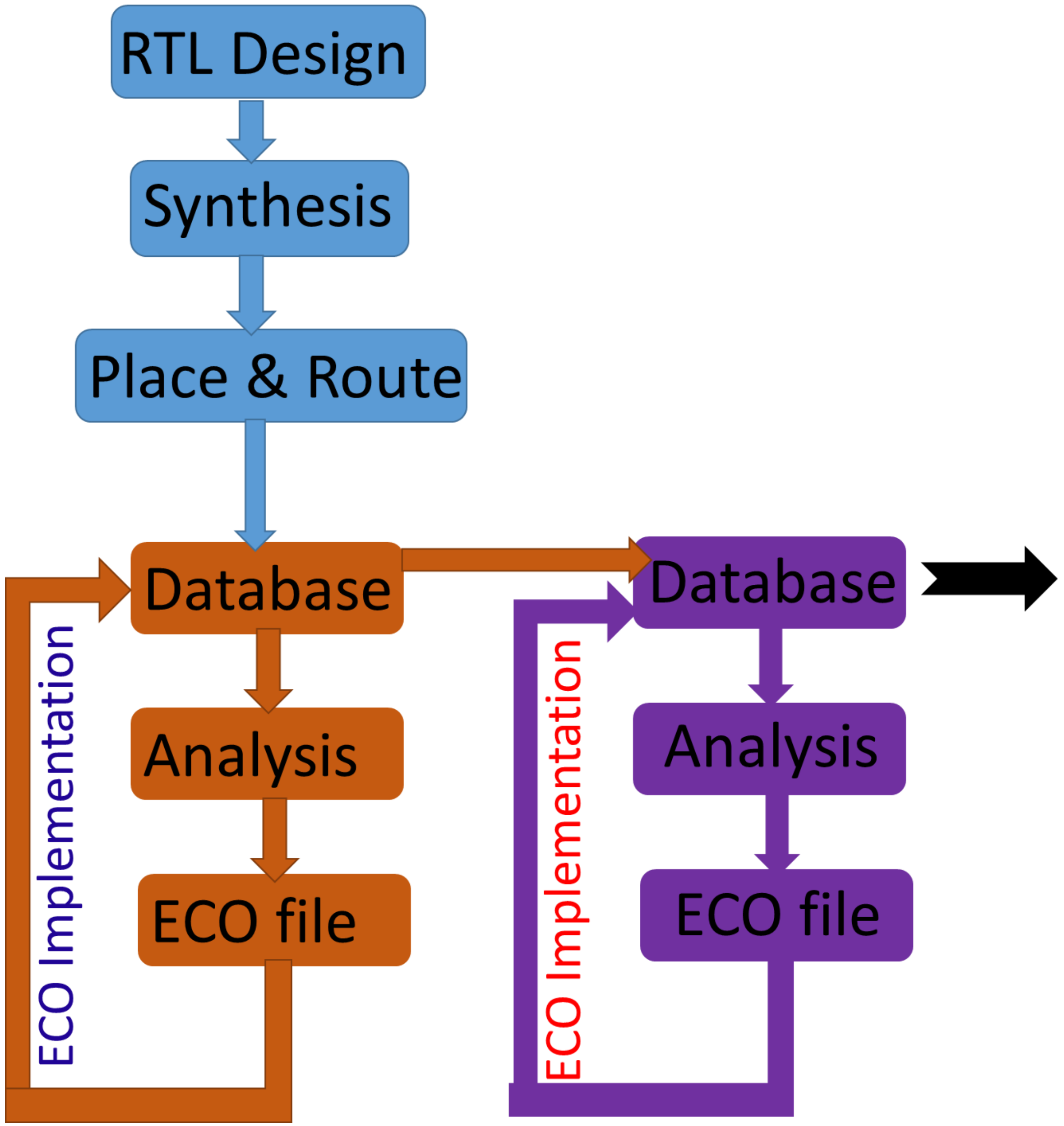
- Routing connects placed cells and pins following Design Rule Checks (DRC).
- Four-step process: *global route*, *track assignment*, *detail routing*, search & repair.
- Goals: minimal wirelength, via count, area footprint, and zero DRC violations.
- Post-routing, check for shorts, opens, antenna violations, and congestion.
- Routing tracks aligned to metal layers; proper pitch selection is critical.

ECO (Engineering Change Order)

The process of sending a clean layout file in form of gds/oasis to the foundry for fabrication after passing all the checks set by the foundry is termed as **tapeout**. But before the tapeout there might be many sleepless nights which Physical Design engineers / Signoff engineers spend to close the design. There are many signoffs like physical signoff, timing signoff and IR signoff which we need to get a closer state after which our layout is ready to send the foundry. And all these final achievements are done in the ECO (*Engineering Change Order*) phase.

ECO Implementation

In the ECO cycle, we perform various analysis one by one for every check which we need to close but not closed till PnR stage. There are specialized signoff tools that help us to analyze the issue and also suggest the changes we need to do in order to close the issue. The suggested change is captured in an ECO file. Once we generate the ECO file for the fixes, we implement that on the PnR database on which we have performed the analysis. After the implementation of ECO file, we save the updated database which we carry forward for the next ECO generation and implementation. We repeat this ECO cycle for every open issue and close one by one all issues. There are chances that we might need multiple ECO cycle to close a single issue.

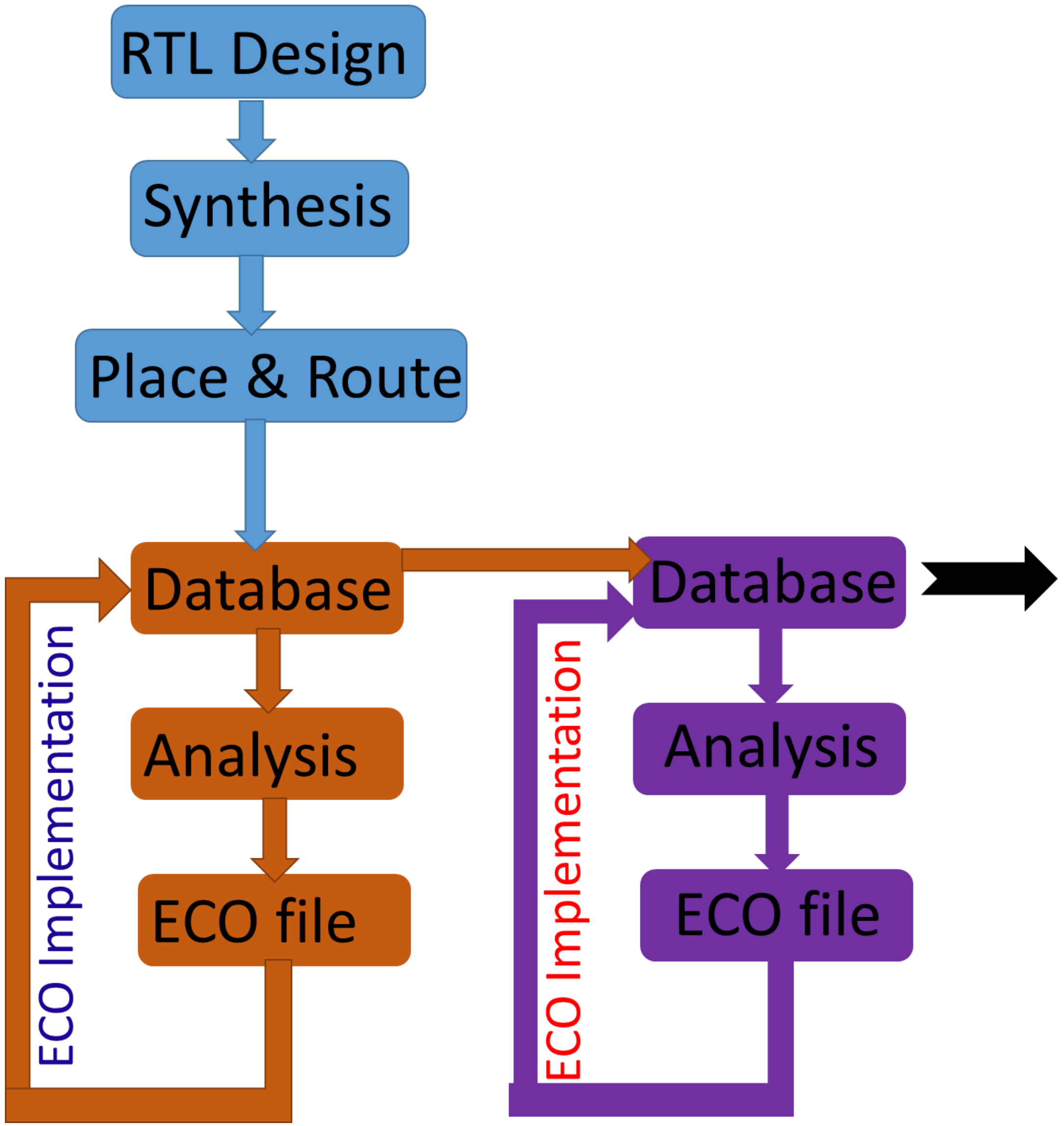


ECO Implementation

Signoff tools are very specialized tools to perform the analysis for a particular issue thoroughly and also have the capability to generate the ECO file for the fixes. We have various types of signoff tools as per the issue like timing signoff tool, Physical Signoff tool and IR signoff tools.

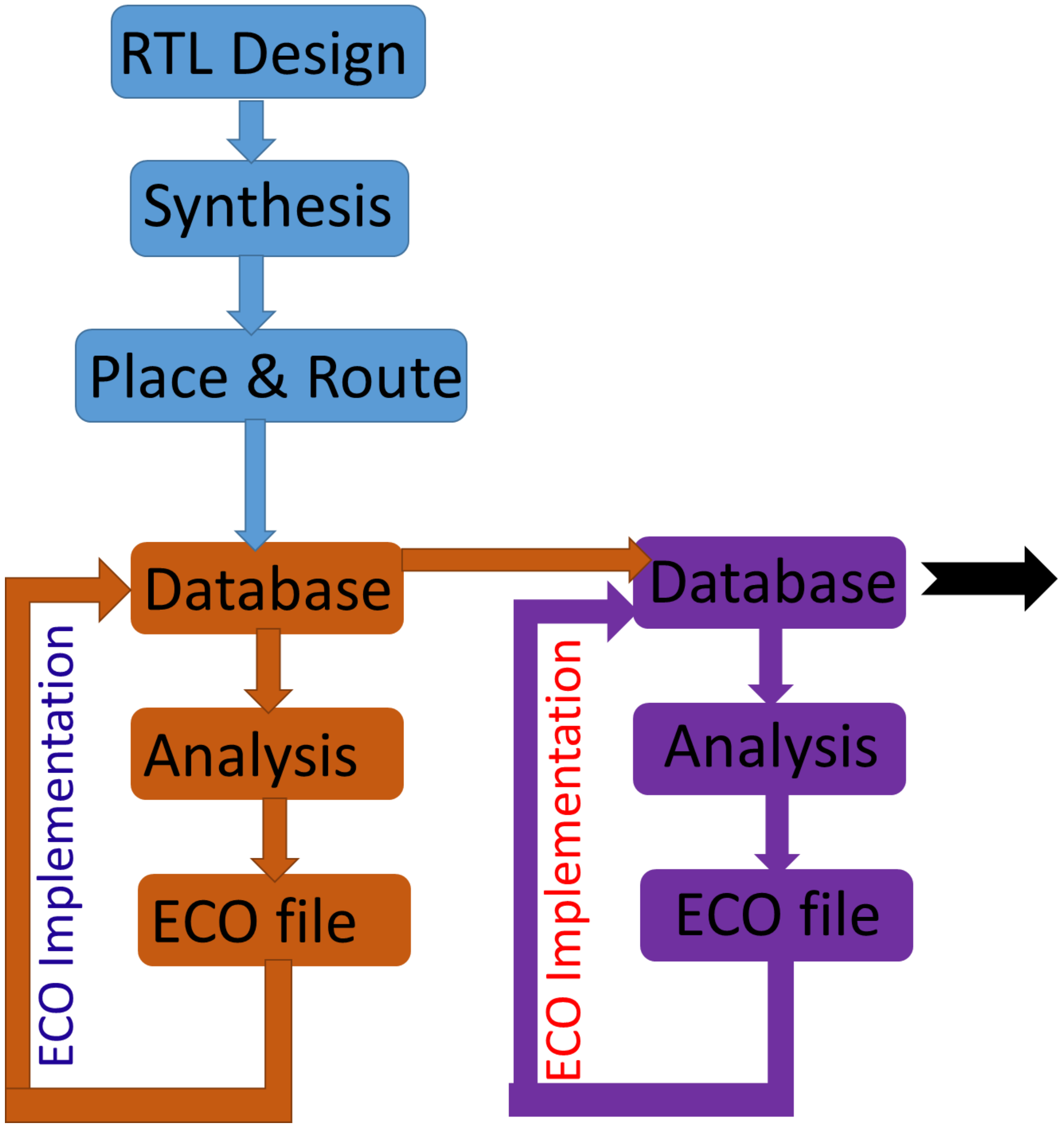
ECO file contains a series of changes required in the form of PnR tool command for fixing the issue. Based on the analysis, sometimes we generate the eco file from the signoff tool itself or sometimes we create our own eco file.

For example, to fix the setup timing, we need to upsize the combinational cells or need to convert them to lower V_{th} cells. In the case of a few hundred violating paths, these conversions of cells might be in thousands or more. So, signoff tool will generate commands for each cell that need to be changed and write in a file that we call eco file. Later we source these file in PnR tool followed by refine placement and ECO route.



ECO Implementation

Once we have the solution of any issue in the form of ECO file, we need to load the database which was used to generate the ECO file and source the eco file. ECO implementation is generally done in the batch mode of the tool. We need to delete the fillers before sourcing the eco file. Once the ECO file is sourced, all the required changes is done. Now there might be a change in the size of cells or addition/deletion of some cells so we need to do refine placement to followed by the ECO route. These two steps will take care of any overlap of cells and routing of cells. The updated database need to be saved for next stage of ECO or the final database in case of all issues are fixed.



ECO Flow in Physical Design

ECO begins after logic synthesis and often post-place and route, during or after timing closure signoff stages. The flow typically includes:

- Capturing ECO specifications (functional fixes or timing fixes).
- Making changes at the gate-level netlist directly corresponding to RTL changes or timing adjustments.
- Updating placement and routing locally around the changes to minimize disruption.
- Re-running local re-synthesis only on affected portions.
- Re-checking timing closure with the new changes.
- Verifying ECO correctness using functional verification and formal equivalence checking with original RTL.
- Finalizing updated layout and physical design data before tapeout.

EDA tools help to manage ECO by enabling localized re-synthesis, minimizing changes only to impacted modules or nets, and supporting incremental timing analysis to avoid full re-implementation. Physical-aware ECO routing minimizes metal mask changes, reducing cost and turnaround time.

Scenarios Requiring ECOs

- Functional bug fixes discovered after synthesis or layout.
- Customer-driven feature or specification changes late in the flow.
- Timing closure fixes addressing hold/setup violations.
- Power or clock network fixes and optimizations.
- Corrections of verification or formal checking failures after implementation.

Typical Design Changes in ECOs

- Functional ECOs: Gate-level netlist modifications to fix logical behavior or add minor feature updates without a full resynthesis of the chip.

- Fix-based ECOs: Changes targeting timing violations such as buffer insertion, gate sizing, upsizing drivers, rerouting nets, or clock tree resynthesis.

- Power/Clock ECOs: Adjustments to power gating cells, clock gating, or clock tree network restructuring to meet power or timing goals.

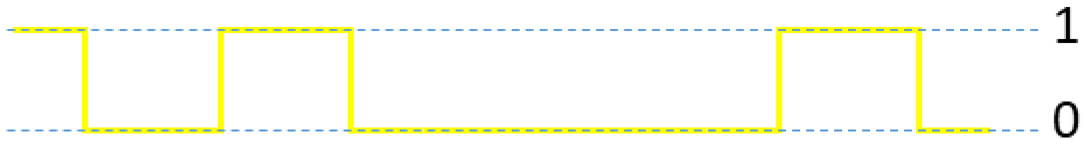
- Metal-mask ECOs: Minimal routing or metal layer changes post-layout to fix electrical issues or timing without changing standard cell placement significantly.

ECO managing Re-Synthesis, Timing Closure & Layout Changes

- ECO tools perform incremental synthesis and optimization limited to affected regions, preserving the remaining layout and constraints.
- Timing closure is maintained by selectively re-optimizing critical paths impacted by ECO, often using late-stage optimization techniques like clock tree resynthesis and local buffer insertion.
- Incremental place-and-route or ECO routing steps update metal layers or local wiring with minimal impact to surrounding structures.
- Formal equivalence checking ensures that ECO logic changes correspond exactly to RTL fixes and no unintended changes propagate.
- Design constraints are reused or selectively updated to maintain timing, power, and design rule adherence without full redesign cycles.
- ECO flow attempts to minimize mask changes (pre-mask ECO) or apply metal-only routing changes (post-mask ECO), preserving manufacturing schedules.

Signal Integrity

Signal could be defined as information in the form of wave/impulse which is used for communication between two points. In digital form, it is either 1 (high) or 0 (Low)



By definition, the integrity means "complete or unimpaired" or we can say that maintaining the actual form of anything over time without any distortion.

So, *signal integrity* could be defined as replication of the entire signal while transmitting from one point to another without any distortion in its quality.

or

***Signal Integrity* is the ability of an electrical signal to carry information reliably and resist the effects of high-frequency electromagnetic interference from nearby signals**.

Signal Integrity Issues

Signal Integrity may be affected by various reasons, but major reasons are:

1. *Crosstalk (Delay and Noise)*
2. *Ground bounce*
3. *IR Drop*
4. *Antenna effect*
5. *Electromigration*

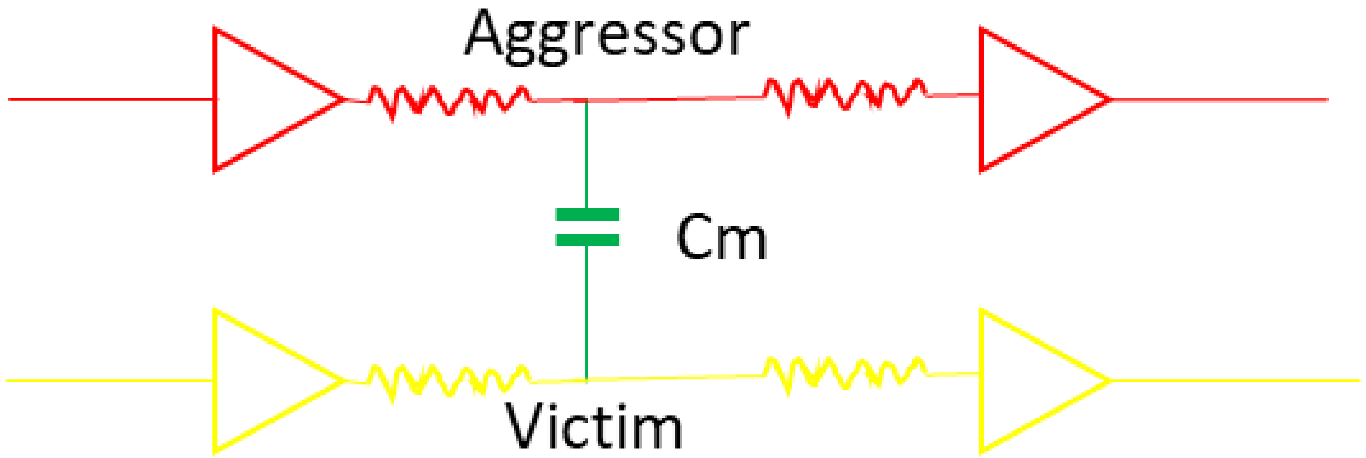
Signal Integrity Issues

What is Crosstalk?

Crosstalk could be defined as a phenomenon in which logic transmitted in one net creates undesired effects on its neighboring nets. Or in another world, we can say switching of a signal in one net can interfere in the neighboring net, which is called crosstalk.

Signal Integrity Issues

When a signal switches, it may affect the voltage waveform of a neighboring net. The switching net is typically identified as the "aggressor" and the affected net is the "victim."



Crosstalk is a very severe effect especially in **lower technology node** and **high-speed circuits** and it could be one of the main reason of **chip failure.**

Signal Integrity Issues

Crosstalk mechanism

Crosstalk occurs via two mechanisms:

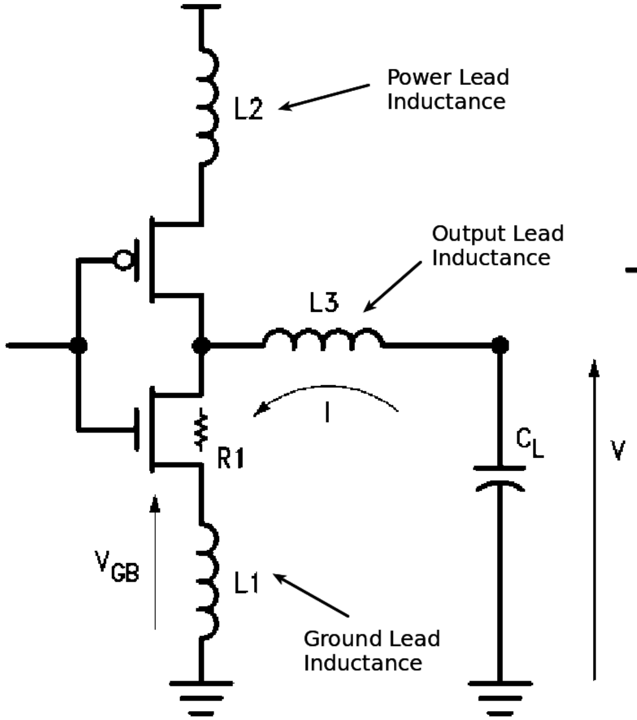
1. Inductive Crosstalk
2. Electrostatic crosstalk

Inductive crosstalk occurs due to **mutual inductance** between two nets. A varying current in a net creates a varying magnetic field around the net. A varying magnetic field can either radiate energy by launching radio frequency waves or it can couple to adjacent nets. Such coupling of the magnetic field is called inductive crosstalk.

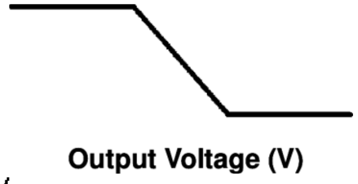
Electrostatic crosstalk occurs due to **mutual capacitance** between two nets. The electric voltage in a net creates an electric field around it. If the electric field is changing, It can either radiate the Radio waves or can couple capacitively to the adjacent net. Such coupling of the electric field is called electrostatic crosstalk.

Electrostatic Crosstalk is significant & problematic than Inductive crosstalk.

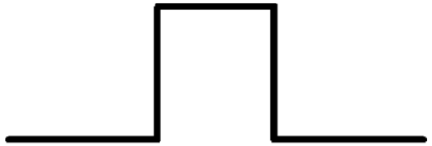
Signal Integrity Issues



(a)

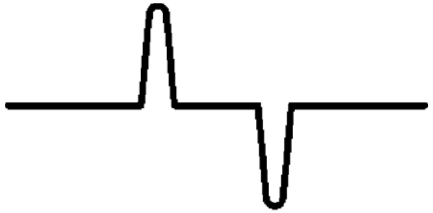


(b)



$I = -C_L \cdot (\Delta V/\Delta t)$

(c)



$V_{GB} = L \cdot (\Delta I/\Delta t)$

(d)

Fig. (a): Simple model to understand ground bounce

The inverter represents a driver in the lead frame of the package driving a load C_L. The inductances L1, L2, L3 represent the intrinsic inductances in the ground lead, power lead and output lead of the package receptively. R1 represents the output impedance of the driver. Note that ideally, we would want inductances L1, L2, L3 to be absent. The driver is now given a rising step input, so that its output falls gradually. Figures (b), (c) and (d) show output waveforms for V , I and V_{GB} against time (symbols are marked in Figure 4.1.1). The output voltage V slew rate is dependent on C_L, L3, L1 and the pull-down transistor. For simplicity, the output voltage V is shown to fall at a constant rate in Figure (b). As the output voltage V falls, the capacitor starts to discharge and generates a current I as shown in Figure (c). This discharging current goes to the ground through the inductances L3 and L1. The voltage waveform across the ground lead inductance V_{GB} is shown in Figure (d), clearly shows that the internal device ground is not same as the external system ground at all times or in other words, the internal device *ground bounces*.

Signal Integrity Issues

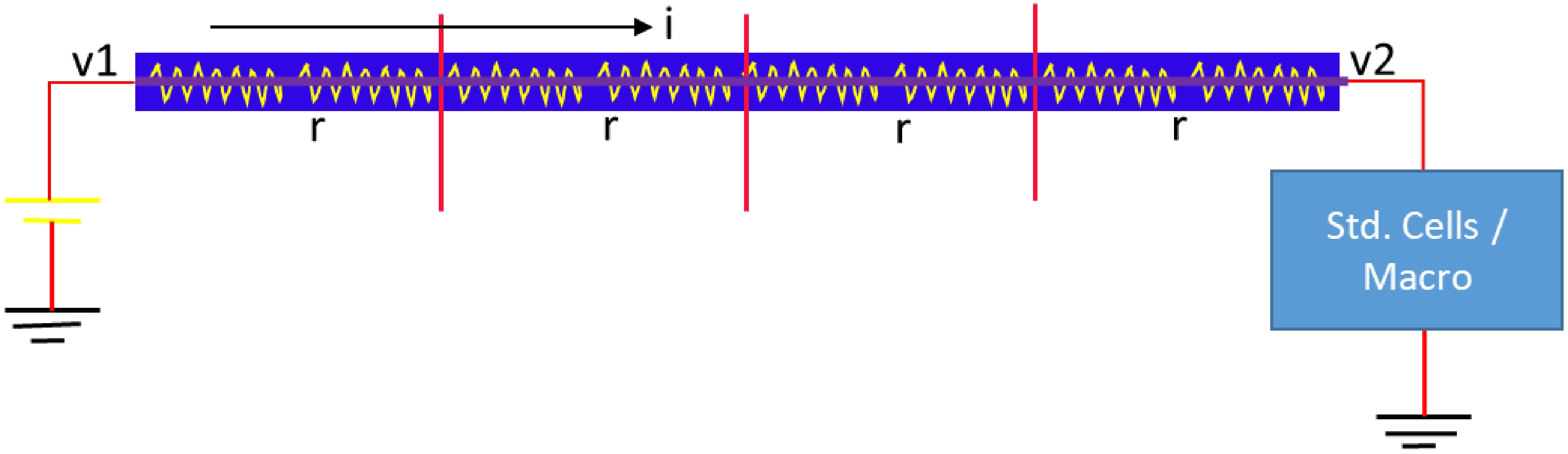
Ground Bounce

A critical phenomenon that arises predominantly in high-speed digital circuits, particularly where rapid signal switching occurs. It is crucial to understand that *ground bounce* is the result of changes in the return current path caused by the inductive effects of circuit layout and design. When signals switch from low to high states, they can generate a sudden demand for current, which forces the return current to traverse a longer path. This can lead to voltage discrepancies at the ground reference point, causing the signal integrity to suffer significantly.

Signal Integrity Issues

IR Drop

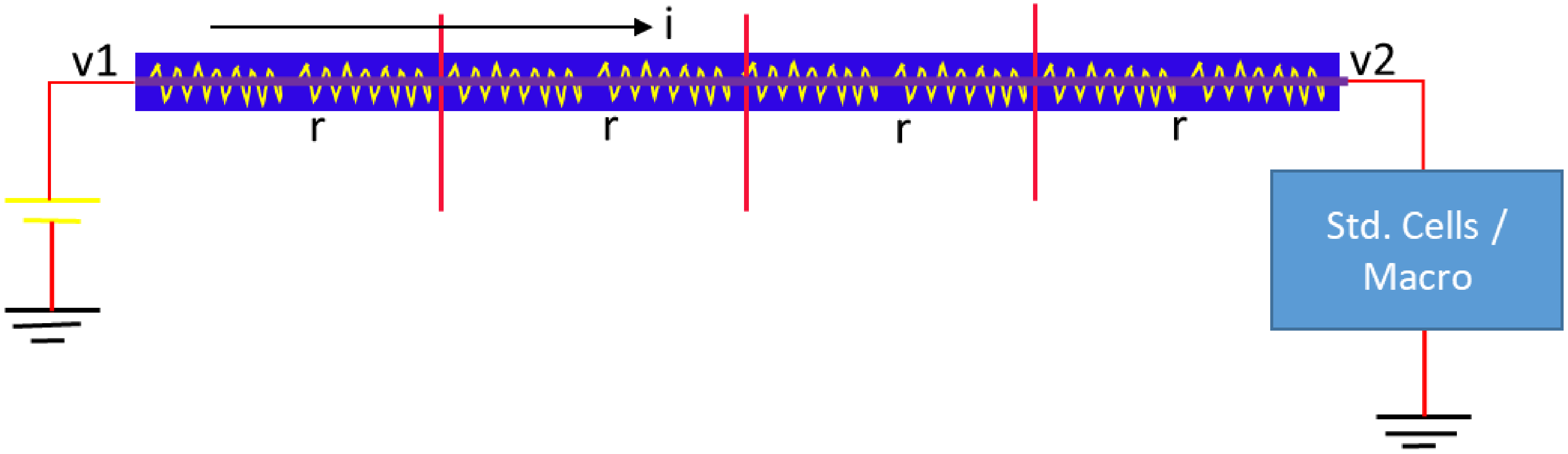
The power supply (VDD and VSS) in a chip is uniformly distributed through the metal rails and stripes which is called Power Delivery Network (PDN) or power grid. Each metal layers used in PDN has finite resistivity. When current flow through the power delivery network, a part of the applied voltage will be dropped in PDN as per the Ohm's law. The amount of voltage drop will be V = I×R, which is called the *IR drop*.



Signal Integrity Issues

IR Drop

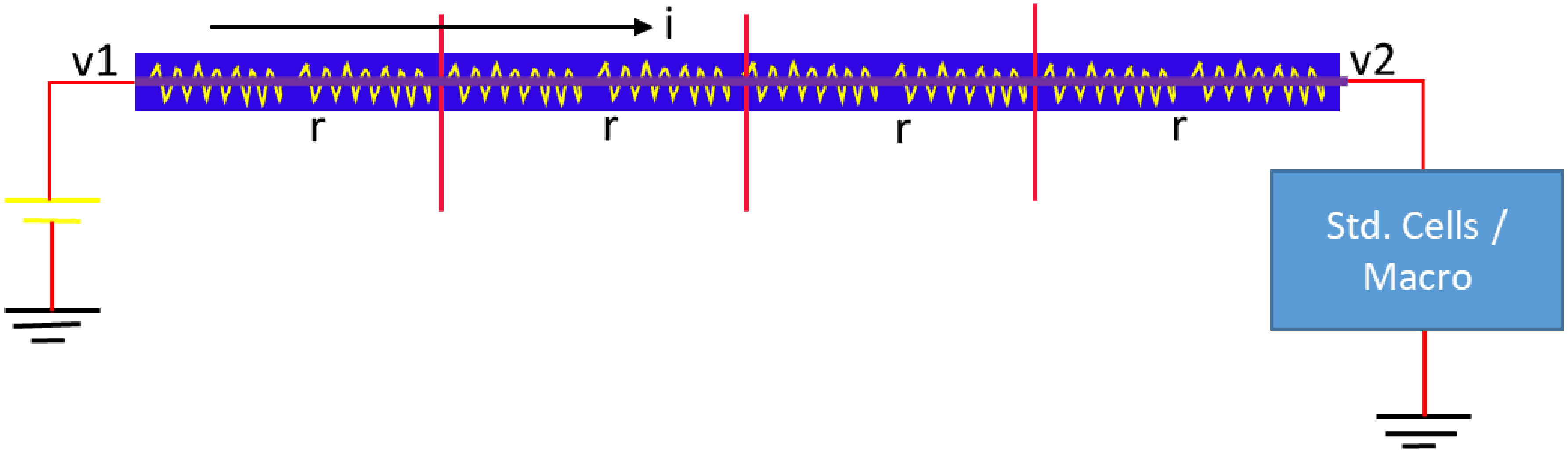
The below figure shows the IR drop in the *Power net*. Any metal net can be assumed as a combination of small R and C. If the resistivity of metal wire is high or the amount of current following through the power net is high, A significant amount of voltage may be dropped in the power delivery network which will cause a lesser amount of voltage available to the standard cells than the actual amount of voltage applied.



Signal Integrity Issues

IR Drop

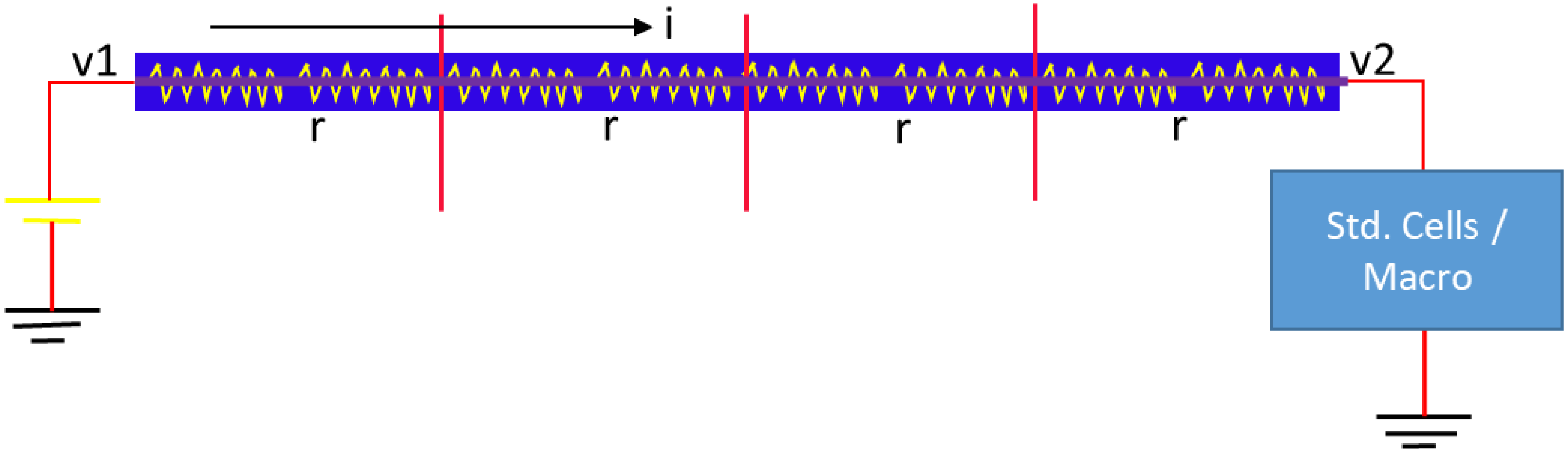
If V1 voltage is applied at the power port and current I is following in a particular net which has total resistance R, then the voltage available (V2) to the other end for the standard cell will be $\boldsymbol{V2 = V1 - I \times R}$



Signal Integrity Issues

IR Drop

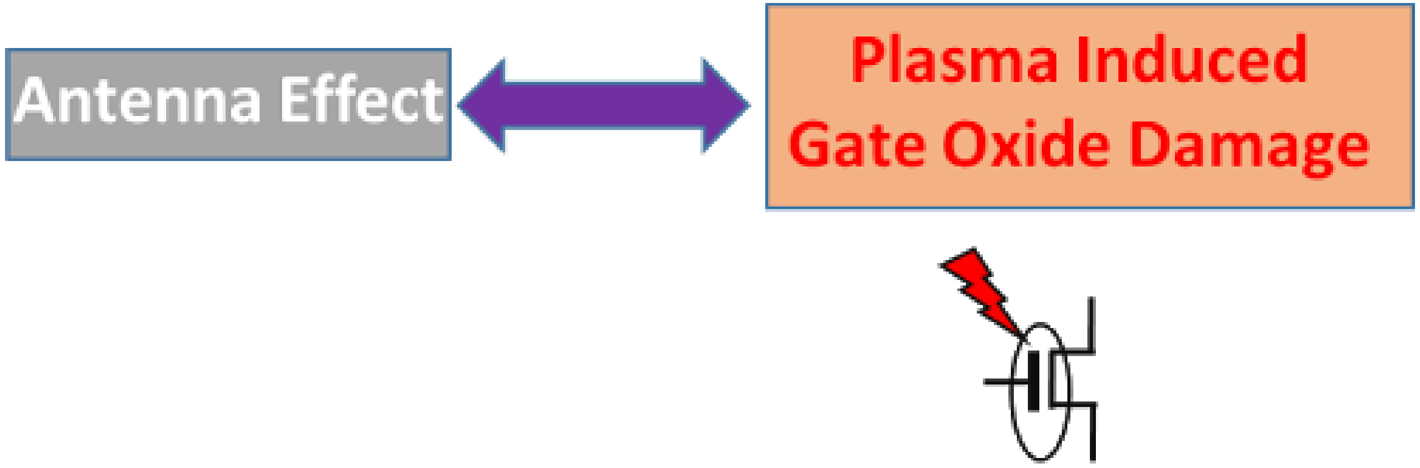
Standard cells or macros sometimes do not get the minimum operating voltage which is required to operate them due to IR drop in power delivery network even the application of sufficient voltage in the power port. Voltage drop in the power delivery network before reaching the standard cells is called *IR drop*.



Signal Integrity Issues

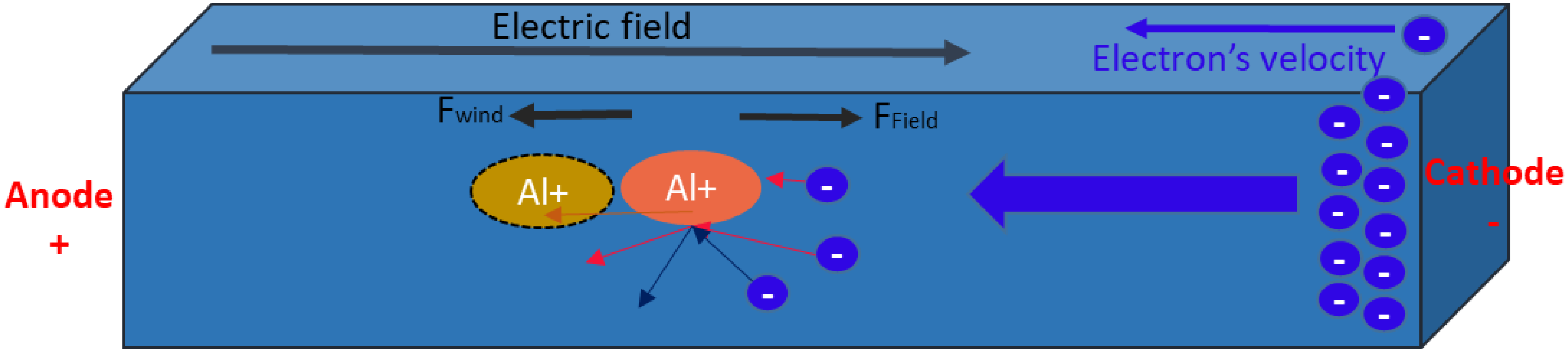
Antenna Effect

The term Antenna Effect might not give you the right intuition about the actual effect it may lead you to think about electromagnetic radiation or transmitter-receiver concepts but here the case is different. So, it has another popular name which is called *"Plasma Induced Gate Oxide Damage"* which provides the right intuition about the effect. As this name itself indicates that this is an effect caused by the *Gate Oxide Damage* due to the *Plasma Etching* process during the fabrication process.



Signal Integrity Issues

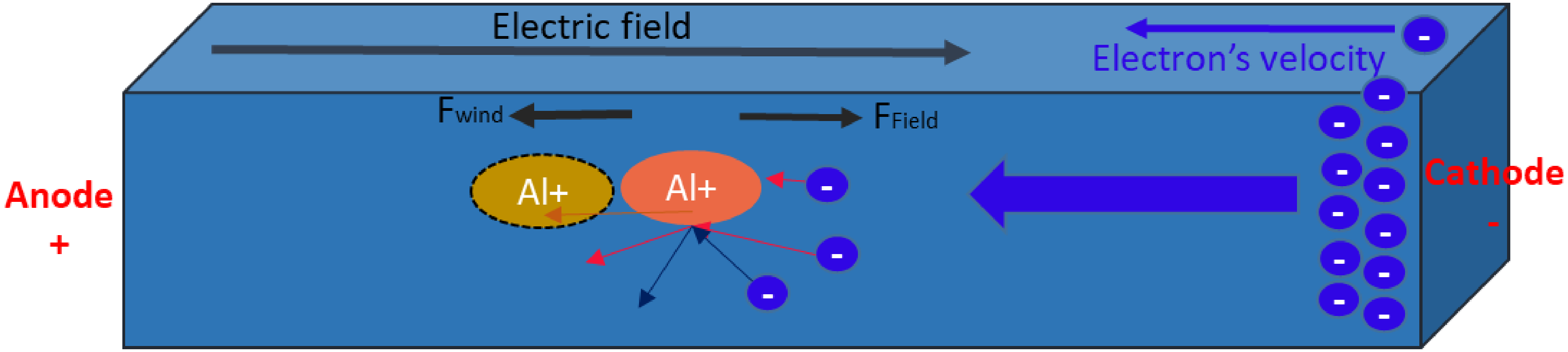
Electromigration



The above figure shows the phenomenon of electromigration effect. A potential difference is applied across a metal interconnect which setups an electric field from anode side to cathode side as shown. This electric field causes to move the electron in the opposite direction of the electric field. This momentum of electron cause flow of current in the electron. These moving electrons have momentum and when it collides with the metal ions the metal ions feel two forces in the opposite direction as shown in the figure. One force and due to electric field and other is due to strike of electrons wind. If the current density is high the force due to electron wind is greater than the force due to the electric field.

Signal Integrity Issues

Electromigration



Depending on the current density, the subjected metal ion started drifting in the opposite direction of the electric field. If the current density is high, the interconnect may get affected of EM instantly or some times the effect may come after months/years of operation depending on current density. So, the reliability of ASIC will depend upon this EM effect.

Signal Integrity Issues

Electromigration

When a high current density passes through a metal interconnect, the momentum of current-carrying electrons may get transferred to the metal ions during the collision between them. Due to the momentum transfer, the metal ions may get drifted in the direction of motion of electrons. Such drift of metal ions from its original position is called the electromigration effect. The current density J is defined as the current following per unit cross-section area. J = I/A, where I is the current and A is the cross-section of the area of interconnect. As the technology node shrinks, the cross-sectional area of the metal interconnects also shrinks and the current density increases in great extent in the lower node. Electromigration has been a problem since 90 nm technology node or even earlier but it gets worse in lower technology node 28nm or lower node.